Larry Nguyen

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Education

Bachelor of Science in Electrical Engineering (B.S.) Anticipated Graduation: Dec. 2027
University of Maryland, College Park, MD *GPA: 3.01 | Upper-Level EE GPA: 3.58*

Relevant Coursework: Advanced Analog and Digital Electronics, Digital Circuits and Systems Lab, Digital Logic Design, Electric Circuits, Electromagnetics, Introduction to Micro and Nanoelectronics, Micro and Nanoelectronics Lab, Signal & System Theory

Technical Projects

Adjustable Linear Bench Power Supply Jul. 2026 – Present
Personal Project

- Designed a 0–15V, 1.2A linear power supply with rectification, filtering, relay protection, and regulation stages; sized a 2200 µF filter capacitor for a $<$50 mV ripple target
- Simulated in LTspice (15V output, current limiting over a 0.1–1A range) before layout; completed PCB layout and fabrication
- Sized the pass-transistor heatsink from a worst-case ~28W dissipation calculation and designed current-limiting protection via a 0.5 Ω sense resistor
- Diagnosed and resolved a hard fault on regulator output by tracing the failure through the regulation chain

DC-DC Buck Converter with PID Control Jun. 2026 – Jul. 2026
Personal Collaborative Project

- Designed a 12V-to-5V, 2A buck converter switching at 101 kHz, driven by an LTC1693-1 gate driver and IRLZ44N MOSFET
- Validated switching waveform on an oscilloscope against simulation; calculated output ripple of ~78 mV with a 10 µF output capacitor
- Diagnosed a load-resistor (R1/R3) footprint mismatch and corrected it by mounting the resistors off-board

Electronics Lead – Crushin' It Sept. 2023 – Dec. 2023
Semester Project

- Led electronics for a six-person team's autonomous vehicle that located, measured, and approached a target; awarded Most Creative OTV among six teams
- Wired color and Time-of-Flight sensors and three servos to an Arduino Mega, routing I^2C and PWM signals

Arcade Machine Restoration & Modernization Project Dec. 2025
Personal Project

- Upgraded a vintage arcade cabinet to support 60,000+ retro games via a Raspberry Pi 5 running Batocera Linux
- Removed obsolete hardware and rewired 15 LED-illuminated buttons/joysticks to a USB encoder board on the Pi
- Wired the Raspberry Pi to the monitor's controller board over HDMI, replacing the cabinet's stock video path

Other Experience

Server & Busser Jun. 2025 – Present
Pho An & Grill – Silver Spring, MD

- Served up to 15 tables (60 customers) during busy shifts, 3 shifts per week, and helped maintain smooth operations
- Trained 8 new employees on restaurant procedures and day-to-day service

Externship | Boiler and Chiller Operations Sept. 2023 – May 2026
Navy Yard & UMBC Mechanical Rooms — Washington, D.C. / Baltimore, MD

- Shadowed boiler/chiller operators across two facilities (100+ hours over 20+ visits): contactor replacement, wire pulling, and 0V diagnostics
- Troubleshot 2 VFDs, plus relays and contactors, in mechanical cooling/heating equipment using multimeters and non-contact voltage testers

Skills

Hardware: Arduino, ESP32, KiCad (schematic and PCB layout), LTspice, Oscilloscope, Raspberry Pi

Software: Fusion 360, GitHub, MATLAB, MIPS Assembly, Verilog